

HARINARAYANAN P

+91 6374369428 | harinarayananpari@gmail.com | github.com/Harinarayanan-Pari

[linkedin.com/in/harinarayanan-pari](https://www.linkedin.com/in/harinarayanan-pari) | <https://harinarayanan-portfolio-ai.web.app/>

SUMMARY

Software Engineer with **2+ years of experience** building AI-powered automation, full-stack applications, and document intelligence systems. Proven track record of designing end-to-end AI systems, workflow automation, and cloud-native applications in fast-paced startup environments. Expertise in Python, FastAPI, React.js, OpenAI/Gemini APIs, RAG pipelines, MongoDB, PostgreSQL, Elasticsearch, AWS, Azure, and scalable backend architectures. Comfortable across the full stack — from designing AI pipelines and REST APIs to building responsive frontends — making me an effective contributor whether the role demands backend depth, full-stack breadth, or AI-first engineering.

WORK EXPERIENCE

Software Developer – Fullstack & AI Systems, Finkraft.ai - Bengaluru

Apr 2024 – Apr 2026

- **Built an end-to-end AI-powered document intelligence platform** with Microservices using OpenAI, Gemini, Vertex AI, RAG, and Flask that **processed 200,000+ invoices and enabled ₹250+ crore GST Value reconciliation for Fortune 500 clients**, reducing manual effort by 40%, increasing throughput by 3–5x, and contributing to customer growth from 10 to **100+ enterprise clients**.
- **Architected and developed an Elasticsearch-based fuzzy matching and reconciliation engine** from scratch for invoice, GSTR-2B, and booking datasets, **improving matching accuracy by 40%** and enabling scalable client-facing reconciliation workflows.
- Led **R&D and implementation of five generations** of document parsing and data matching pipelines, continuously improving extraction accuracy, scalability, and processing reliability.
- **Developed an AI-powered chatbot using OpenAI APIs** and **built a cloud-agnostic bulk file download** solution integrating AWS S3, Azure Blob Storage, asset links, automated ZIP generation, and Zoho Mail-based delivery workflows.
- Developed an **automated vendor follow-up email system** using SendGrid and email scraping system, streamlining invoice collection workflows and reducing manual communication efforts.
- **Developed full-stack web applications for vendor invoice collection**, data visualization, and operational dashboards using React.js and Flask.
- **Designed ETL pipelines, validation frameworks, and backend services** using FastAPI, MongoDB, and PostgreSQL to ensure reliable data processing at scale.
- Led **migration from AWS S3 to Azure Blob Storage as a cost-optimization** initiative, reducing infrastructure expenses while implementing high-volume batch processing systems with structured logging and observability.
- Leveraged **AI-assisted development tools** including Cursor, GitHub Copilot, and Antigravity.
- **Converted from Full-stack Intern to Full-Time based on strong performance and contributions** to core product development initiatives.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript
- **AI & Generative AI:** OpenAI API, Gemini API, Vertex AI, Prompt Engineering, RAG Pipelines, LLM Application Development, Structured Output Parsing, AI Automation
- **Backend Technologies:** FastAPI, Django, Flask, REST APIs, Microservices
- **Frontend Technologies:** React.js, HTML5, CSS3
- **Search & Retrieval:** Elasticsearch, Vector Search, Semantic Search, Index Design, Fuzzy Matching, Query Optimization
- **Databases:** MongoDB, PostgreSQL
- **Cloud & Devops :** AWS S3, Azure Blob Storage and Docker
- **Software Engineering:** OOP, Data Structures & Algorithms (DSA), System Design, Design Patterns, Scalable Backend Development
- **Tools:** Git, GitHub, GitHub Copilot, Cursor, Structured Logging

KEY PROJECTS

- AI-Powered Invoice Parsing & Matching Service | Python, LLM(OpenAI, Gemini, Vertex AI), RAG, Vector Search, Elasticsearch
Impact : Automated invoice extraction, validation, and GST reconciliation.
- AI Customer Support Chatbot with RAG | LLM(OpenAI), RAG, Vector Search, React.js, Flask
Impact : Delivered 24/7 AI-powered customer support through human-like conversational assistance.
- Intelligent Reconciliation Engine | Elasticsearch, Python, Fuzzy Matching
Impact : Automated invoice, GSTR-2B, and booking data reconciliation.
- Cloud-Agnostic Bulk File Download Platform | AWS, Azure, FastAPI, Zoho mail, Flask, React.js
Impact : Automated bulk document downloads and ZIP-based delivery.
- Full-Stack Vendor Invoice Collection Portal | React.js, Flask, LLM
Impact : Streamlined vendor invoice collection and operational workflows.
- Auto-Followup Mailing & Email Scraping System | SendGrid, Python
Impact : Automated vendor follow-ups and invoice retrieval from emails.

EDUCATION

Bachelor of Engineering – Mechanical Engineering

Rajiv Gandhi College of Engineering

Jun 2014 – Apr 2018

Kanchipuram,TN